Intro to Bayesian

UBCO MDS — DATA 582

Statistical Paradigms

- The field of statistics is often broken down into two different (sometimes argued as competing, but more accurately described as orthogonal) philosophies for inference

1. Classical or frequentist statistics
2. Bayesian statistics

- Both inferential methods have the same starting point: we assume some family of probability distributions F where

$$\mathcal{F} = \{f_\theta(x); \quad x \in \mathcal{X}, \quad \theta \in \Omega\}$$

with x being observed data from sample space $\mathcal{X}$, θ is some unobserved value from parameter space Ω, and our goal is infer something about θ from observing x.

Classical/Frequentist

- The frequentist framework is probably where the vast majority of you have spent most (or even all) of your statistical education.
- We assume the parameter θ is some fixed, but unknown, constant and we set out to consider how we might estimate it given a sample.
- A frequentist typically views this search through considering how their estimator for θ would behave with repeated sampling from $\mathcal{X}$.

Classical/Frequentist

- This is where the ideas surrounding sampling distributions, and their importance to statistical inference, originates.
- Consider the typical intro stats examples where $X \sim N(\mu, 8)$
- We drive home that $\bar{X}$ is, itself, a random variable...and if we consider the possible samples of size n we could observe, we find that $\bar{X} \sim N(\mu, 8/n)$...
- Therefore if we compute an observed $\bar{x}$ from a single sample of size n, we assign it the variability $8/n$ from the random variable $\bar{X}$
- That lets us do hypothesis tests and CIs for the parameter μ.

Classical/Frequentist

- You've likely seen this line of reasoning so often that it's almost second nature, and hard to see where the flaws might be
- One common flaw is that the domain $\mathcal{X}$ can be misspecified
- Example: I assume operating pizza oven temperatures (fahrenheit) arise normally $X \sim N(\mu, 100^2)$ and I measure temperature 10 times using my temperature gun. I later notice my temperature gun can only measure between -40 and 1200 degrees (and would provide those min, max, when exceeded).
- My observed $\bar{x}$ no longer technically has those properties I'm used to for estimating μ even if my observed sample never actually strayed into the truncated areas

Classical/Frequentist

- Why? Because my theoretical repeated samples from $\mathcal{X}$ are being pulled from f, which is on the domain $-\infty, \infty$. And those are what dictate the properties of $\bar{X}$, which no longer theoretically apply to my observed sample $\bar{x}$.

- In practice, most would generally accept this type of misspecification as an approximation error.

- But from a mathematical/theoretical standpoint, it is basically ubiquitous and therefore somewhat troubling...

Bayesian

- The Bayesian approach simply bakes in one more assumption: that we can quantify some prior knowledge about θ.
- The easiest way to conceptualize this is moving from thinking of θ as a fixed, but unknown, constant to thinking of it as a random variable.
- Approaching statistics from this standpoint shifts both the philosophy and methodology surrounding inference.
- Let's start with a quick review of Bayes Theorem (on board).

Bayesian - Discrete

- We have always considered a sample $X_1, \dots, X_n$ to be a set of random variables, and $\mathbf{x} = x_1, \dots, x_n$ and observed sample, but if we now consider θ to be a random variable...then Bayes Theorem dictates

$$P(\theta \mid \mathbf{x}) = \frac{P(\mathbf{x} \mid \theta)P(\theta)}{P(\mathbf{x})}$$

Bayesian - Continuous

- Similarly for continuous we could show

$$h(\theta \mid \mathbf{x}) = \frac{f(\mathbf{x} \mid \theta)g(\theta)}{f(\mathbf{x})}$$

- We call $h(\theta \mid \mathbf{x})$ the posterior distribution of θ. Inferentially, it fully describes θ under the assumptions we've made.
- $g(\theta)$ the prior distribution of θ.
- $f(\mathbf{x} \mid \theta)$ you've seen many times before: it's the likelihood function (which we often maximize to find estimators for θ).
- $f(\mathbf{x})$ is a weird one. Technically, it's some sort of marginal likelihood where θ is integrated out: $\int_\Omega f(\mathbf{x} \mid \theta)g(\theta)d\theta$

Bayesian - Continuous

- In practice, since we know that $h(\theta \mid \mathbf{x})$ is a probability distribution, and since the observed $\mathbf{x}$ is fixed, we think of $f(\mathbf{x})$ as merely a normalizing constant.
- In fact, we will regularly ignore it via the relationship

$$h(\theta \mid \mathbf{x}) \propto f(\mathbf{x} \mid \theta)g(\theta)$$

- The posterior distribution of θ depends on the observed data (through the likelihood) and the prior distribution we imposed on θ.

Bayesian vs Frequentist

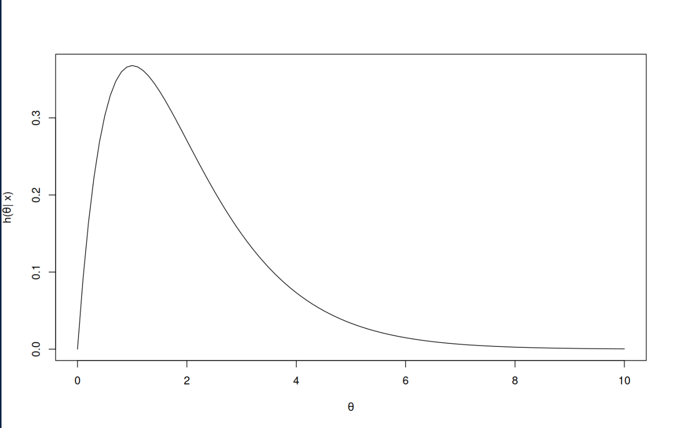
- Bayesians consider the path to inference by assuming variability in the parameters θ. Notably, they consider the observed data $\mathbf{x}$ FIXED.
- Frequentists consider the path to inference by assuming variability in the observed data $\mathbf{x}$. Notably, they consider the parameters θ FIXED.
- They approach the same problem (learning what we can about θ) in orthogonal ways.

Bayesian vs Frequentist

- Returning to our pizza oven example: a Bayesian would not care that the temperature gun was constrained, since the observed data was no different than what would have been observed with a perfect measuring device.
- Since the data is unaffected by the constrained device, the observed likelihood is also unaffected, and our inferential framework in total is therefore unaffected.
- Not only that, but in the pizza oven example, we probably could propose a sensible prior distribution for what we would expect to see for μ: the average temperature of an operating pizza oven!

Posterior Distribution

- Pretend we simply had $h(\theta \mid \mathbf{x})$ on hand and that θ is some super important parameter: maybe, the global temperature increase going forward from today that causes humanity's demise.



- What might you do with this info?

Posterior Distribution

- The posterior gives a full description for θ (under the assumptions of your model for $\mathbf{x}$ AND your prior for θ)
- Just like in the frequentist paradigm, we might still want to summarize our knowledge of θ in several fashions.
- What's our single best guess for how many degrees we can allow Earth to increase before we're doomed?
- What about a range of values? How might we build that range? How might we interpret it?

Bayesian

- There will be changes to some of these things in the Bayesian paradigm
- But, it's important to emphasize that our overall tasks are quite similar.
- Learn what we can about θ, and use that information to guide decision making.

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- We will build our knowledge initially through simple examples (univariate distributions, etc).
- Work our way towards interesting computational tricks.
- And eventually return to supervised modeling.
- Do not consider this to be comprehensive coverage of a huge topic in statistics, nor myself to be a leading expert in this paradigm :)


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